

EDUCATION

Nanjing University	Nanjing, China
M.A. in Journalism and Communication, specializing in Computational Communication	2024 – 2027
GPA: 4.00/4.00	
Communication University of China	Beijing, China
B.Eng. in Cyberspace Security, Minor in Computational Communication	2020 – 2024
GPA: 3.64/4.00	

PEER-REVIEWED PUBLICATIONS

[2] **Zhang, Y., & Ma, Z.** (2026). Depressive symptom trajectories around the retirement transition across countries and historical periods [Stage 2 Registered Report]. PsyArXiv. https://doi.org/10.31234/osf.io/6ypwr_v6. (Peer-reviewed and recommended by *Peer Community in Registered Reports*)

[1] **Zhang, Y., & Ma, Z.** (2025). Depression trajectories from mid to late life (50–89 Years): The roles of cohort, multimorbidity status, and national contexts across nine European countries. *Social Science & Medicine*, 385, 118605.

SELECTED WORKS IN PROGRESS

[4] **Zhang, Y., & Ma, Z.** (2026). Psychometric comparability of LLM-based digital twins. SSRN. <https://doi.org/10.2139/ssrn.5967255>. (Revise and resubmit at *Nature Communications*)

[3] **Zhang, Y., & Ma, Z.** (2026). Loneliness from midlife to older adulthood reflects both stable trait and changing circumstance. PsyArXiv. https://osf.io/preprints/psyarxiv/7fc9v_v1

[2] **Zhang, Y., & Ma, Z.** (2026). Digital exclusion among middle-aged and older adults in China: Age–period–cohort evidence from three national surveys, 2011–2022. PsyArXiv. https://osf.io/preprints/psyarxiv/hpv68_v1

[1] Yang, Y., **Zhang, Y., & Ma, Z.** (2026). Early identification of adolescent mental health risk through integrated game-based assessment and large language model psychological inference. (Under review at *The British Journal of Social Work*) Contribution: [Data curation; Formal analysis; Methodology].

RESEARCH EXPERIENCE

Computational Communication Collaboratory, Nanjing University	Nanjing, China
Graduate Student Researcher Supervisor: Dr. Zhihao Ma	2024–Present
<i>Psychological Adaptation and Social Participation Across the Later Life Course</i>	Sep. 2024–Present
Investigated how aging, birth cohorts, retirement transitions, and national contexts shape depression and loneliness using large-scale longitudinal and cross-national surveys.	
Examined digital exclusion and reciprocal associations between internet use and mental health, with attention to socioeconomic and health inequalities.	
Applied advanced longitudinal and latent-variable methods, including HAPC growth-curve models, piecewise multilevel models, trait–state–occasion models, mixture modeling, and RI-CLPM.	
<i>Digital Twin Psychometrics</i>	Aug. 2025–Present
Examined psychometric comparability between LLM-based digital twins and human respondents across psychological, behavioral, and free-text tasks.	
Designed multi-study evaluations using large-scale survey and longitudinal datasets across models and prompting conditions.	
Applied psychometric network analysis, statistical calibration, semantic network analysis, and	

computational text analysis to assess construct validity and human–LLM alignment.

Constructing Old Age: A Cross-National Diachronic Analysis of Aging Narratives Oct. 2025–Present
Analyzed narratives of aging across countries and historical periods using large-scale textual corpora.
Constructed diachronic word embeddings to examine age-related stereotypes and social representations of aging.
Applied semantic association analysis, time-series analysis, and multilevel modeling to identify temporal shifts, cross-national differences, and contextual factors shaping aging narratives.

PEER-REVIEWED CONFERENCE PAPERS

[2] **Zhang, Y., & Ma, Z.** (2026). Associations Between Digital Exclusion and Mental Health in Middle-Aged and Older Adults. *76th Annual Conference of the International Communication Association (ICA)*, Cape Town, South Africa, June 4–8, 2026. Accepted for presentation.
[1] **Zhang, Y., & Ma, Z.** (2025). Tracing Digital Exclusion from Mid to Late Life in China: An Age-Period-Cohort Analysis of Socioeconomic, Regional, and Health-related Disparities, 2010–2022. *111th Annual Convention of the National Communication Association (NCA)*, Denver, CO, United States, November 20–23, 2025. Accepted for presentation.

TEACHING AND PROFESSIONAL EXPERIENCE

Teaching Assistant, Nanjing University
Computational Communication Research Training Program (Graduate course) Fall 2026
Causal Inference (Graduate course) Spring 2026
Applications of LLMs in Digital Humanities (Undergraduate course) Spring 2026
Social Research Methods (Undergraduate course) Fall 2025, Fall 2026
Reporting Intern, Hong Kong, Macao and Taiwan News Desk, Xinhua News Agency May–Aug 2023

SELECTED HONORS & AWARDS

Outstanding Graduate Student Award Dec. 2025
National Scholarship for Graduate Students Nov. 2025
First Prize, Academic Scholarship Nov. 2025
Second Prize, Academic Scholarship Dec. 2024

GRANTS

Distinguished Talent Cultivation Program (2025GYB18) ¥6,000, PI Nov. 2025
Project title: Determinants of Digital Exclusion Among Middle-aged and Older Adults

LANGUAGES & SKILLS
